

Assignment -1

BCAK0002 Data analysis and visualization using excel

Last date of submission:- 16 -April-2026 /17-April-2026

1. How would you describe **Data Visualization**, and in what ways does it assist a Data Engineer in making effective decisions?
2. Highlight the key differences between **SQL (Relational Databases)** and **NoSQL Databases**. Also, mention a practical scenario where each would be preferred.
3. What are the essential guidelines to ensure that a visualization does not mislead viewers? Briefly discuss any three.
4. What do you understand by **data quality and consistency**? Explain why maintaining them becomes difficult when integrating multiple data sources.
5. Distinguish between **Matplotlib** and **Seaborn**. In what situations is Seaborn more advantageous for statistical visualization?
6. If a dataset contains dates in multiple formats, how would you standardize them using **Python/Pandas**? Explain the approach.
7. How does a **Box Plot** differ from a **Violin Plot**? Illustrate the components of a Box Plot with proper labeling.
8. For developing a **real-time price monitoring system**, which processing method would you select—batch or stream? Provide justification.
9. Suggest a suitable visualization approach using **Plotly** to study the relationship between sleep hours and stress levels across age groups.
10. A retail company generates large volumes of daily data. Estimate the storage required over a month and express it in terabytes.
11. While visualizing large-scale geospatial data in cloud platforms like AWS QuickSight, what challenges might arise? How can filters improve clarity?
12. Given a dataset with missing values and extreme outliers, how would you perform **EDA** to prepare it for visualization?
13. Explain how tools like **heatmaps** and **pair plots** can be used to identify relationships between variables.

14. In a system processing events continuously, calculate total events over a day and determine the number of anomalies if a certain percentage are outliers.
15. What is meant by a **Data Lake**? How is it different from a **Data Warehouse**?
16. Which function in **Pandas** is commonly used to manage missing (NaN) values during data cleaning?
17. Discuss the importance of data visualization in the overall data lifecycle. Also, differentiate between batch and stream processing in data collection.
18. Compare **OLTP** and **OLAP** systems. Under what circumstances would NoSQL databases be more suitable than relational databases?
19. Outline the stages of the **ETL process** and explain how the transformation phase affects visualization quality.
20. You have been given a dataset of 2,000 student applications in a spreadsheet.
 - a. Describe how you would use PivotTables and Slicers to create an interactive dashboard that filters applicants by "Course" and "State".
 - b. Which chart type would be most effective for showing the percentage of total enrollments per department?